

Your Paper Title Here

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Abstract

Your abstract text goes here. Keep it concise, focused on the core database or data management challenge, your approach, and key experimental findings.

CCS Concepts

• Information systems → Data management systems.

Keywords

databases, query processing, distributed systems

1 Introduction

2 Related Work

3 Methodology

3.1 Preliminaries

3.2 Authority and Scope Completeness

Dynamically integrating conjunctive sub-queries during federated query processing in discovery-based querying at runtime is difficult due to the unbounded nature of the query execution. To show this, we first introduce a sufficient condition for correctly delegating conjunctive queries in the traditional federated case. Then we will show how deciding whether we can correctly delegate these queries in the discovery-based querying case is impossible when not all sources have been discovered.

First we define our discovery-based federated query setting in Definition 1 and 2.

Definition 1 (Discovery setting). A *discovery setting* is a tuple $W = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc} \rangle$: a universe of sources $\mathbb{D}$, a map $\text{data} : \mathbb{D} \rightarrow 2^T$ giving the graph each one holds, a set of seeds $\mathcal{D}_0 \subseteq \mathbb{D}$, and a *discovery function* disc assigning to every BGP B a map $\text{disc}_B : \mathbb{D} \times 2^T \rightarrow 2^{\mathbb{D}}$, where $\text{disc}_B(d, \text{data}(d))$ are the sources that a retrieved d exposes while B is evaluated.

Definition 2 (Reachable set, reference answer). Iterating discovery from the seeds, let $\mathcal{D}_{i+1} := \mathcal{D}_i \cup \bigcup_{d \in \mathcal{D}_i} \text{disc}_B(d, \text{data}(d))$. The *reachable set* is $\text{reach}_B(W) := \bigcup_{i \geq 0} \mathcal{D}_i$, and the *reference answer* to B is $\llbracket B \rrbracket_{G_{\text{reach}_B(W)}}$. Correctness of a delegation is always measured against this reference answer, never against the universe $\mathbb{D}$: sources that are never reached cannot contribute to it.

Given this definition, we recover the traditional federated query processing setting, in which all sources are known beforehand, by taking $\mathcal{D}_0 = \mathbb{D}$ and $\text{disc}_B \equiv \emptyset$.

Next we define an operator to determine the *relevance* of a source for a query execution setting. Relevance records, for each triple pattern and each source, whether that source produces matches. An example of such an operator is the usage of ASK queries to determine exclusive groups in FedX [2].

Definition 3 (Source relevance). Fix a setting W and a BGP B , and write $\mathcal{D} := \text{reach}_B(W)$ for its reachable set; all that follows is

relative to $\mathcal{D}$. Let $\mathcal{P}$ be the set of triple patterns. We write $\llbracket t \rrbracket_d$ for the evaluation of a triple pattern t over the data of source d . The *relevance indicator* $\Delta : \mathcal{P} \times \mathcal{D} \rightarrow \{0, 1\}$ is

$$\Delta(t, d) = \begin{cases} 1, & \text{if } \llbracket t \rrbracket_d \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases}$$

and $\mathcal{D}_t := \{d \in \mathcal{D} \mid \Delta(t, d) = 1\}$ denotes the *relevant sources* of t .

Thus, relevance means that a data source returns answers for a given triple pattern. Next, we introduce a definition of *correct* delegation of conjunctive queries to a set of sources.

Definition 4 (Delegation). Let B be a BGP, $B' \subseteq B$ the delegated fragment, and $S \subseteq \mathcal{D}$ the (only) sources to which B' is sent as a single conjunctive request, where as in Definition 3, $\mathcal{D} = \text{reach}_B(W)$. Write $G_A := \bigcup_{d \in A} \text{data}(d)$ for $A \subseteq \mathcal{D}$. The delegated plan obtains for B'

$$R := \left(\bigcup_{d \in S} \llbracket B' \rrbracket_{\text{data}(d)} \right) \cup \llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}},$$

and the delegation is *correct* if

$$R \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}} = \llbracket B \rrbracket_{G_{\mathcal{D}}}.$$

This definition follows some of the definitions of federation of FedQPL [1]. Next we show the requirements needed to correctly delegate conjunctive (sub)-queries to federated sources.

Theorem 1 (Correct delegation). *Let W be a federated setting, that is $\text{disc}_B \equiv \emptyset$, so that $\mathcal{D} = \text{reach}_B(W) = \mathcal{D}_0$ is known in advance. Let $t_1, \dots, t_n$ be a set of triple patterns forming a BGP B , $B' \subseteq B$ a sub-BGP, and let $d \in \mathcal{D}$. For d we define $\mathcal{D}' = \mathcal{D} \setminus \{d\}$ the set containing all other sources. Then we can correctly delegate B' to d if $\bigcup_{t_i \in B'} \{d' \in \mathcal{D}' \mid \Delta(t_i, d') = 1\} = \emptyset$*

PROOF. We prove the special case of Definition 4 with $S = \{d\}$, so that $\mathcal{D} \setminus S = \mathcal{D}'$. We assume $B' \neq \emptyset$ and that distinct sources use disjoint sets of blank nodes.

The argument rests on the fact that a *single* triple pattern distributes over a union of sources: for every t and every $A \subseteq \mathcal{D}$,

$$\llbracket t \rrbracket_{G_A} = \bigcup_{e \in A} \llbracket t \rrbracket_{\text{data}(e)}, \quad (1)$$

since matching one pattern involves no join. Note that (1) fails for BGPs, which is exactly what makes delegation non-trivial.

By hypothesis no source other than d is relevant to any pattern of B' , so Definition 3 gives

$$\llbracket t' \rrbracket_{\text{data}(d')} = \emptyset \quad \text{for all } t' \in B', d' \in \mathcal{D}'. \quad (2)$$

Step 1. A BGP is the join of its patterns, and each pattern distributes over the sources by (1):

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \bowtie_{t \in B} \llbracket t \rrbracket_{G_{\mathcal{D}}} = \bowtie_{t \in B} \bigcup_{e \in \mathcal{D}} \llbracket t \rrbracket_{\text{data}(e)}.$$

Step 2. Since $\bowtie$ is associative and commutative, the delegated fragment splits off:

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \left(\bowtie_{t' \in B'} \bigcup_{e \in \mathcal{D}} \llbracket t' \rrbracket_{\text{data}(e)} \right) \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}}.$$

Step 3. As $\mathcal{D} = \mathcal{D}' \uplus \{d\}$, the inner union splits, and by (2) its $\mathcal{D}'$ part is empty:

$$\bigcup_{e \in \mathcal{D}} \llbracket t' \rrbracket_{\text{data}(e)} = \underbrace{\bigcup_{d' \in \mathcal{D}'} \llbracket t' \rrbracket_{\text{data}(d')} \cup \llbracket t' \rrbracket_{\text{data}(d)}}_{= \emptyset} = \llbracket t' \rrbracket_{\text{data}(d)}.$$

Substituting into Step 2 and reassembling the join,

$$\llbracket B \rrbracket_{G_{\mathcal{D}}} = \llbracket B' \rrbracket_{\text{data}(d)} \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}}. \quad (3)$$

Step 4. It remains to identify R from Definition 4. Its second term vanishes, again by (1) and (2) and using $B' \neq \emptyset$:

$$\llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}} = \bigcup_{t' \in B'} \llbracket t' \rrbracket_{\text{data}(d')} = \bigcup_{t' \in B'} \emptyset = \emptyset.$$

With $S = \{d\}$ the first term of R is the single request to d , so $R = \llbracket B' \rrbracket_{\text{data}(d)} \cup \emptyset = \llbracket B' \rrbracket_{\text{data}(d)}$.

Conclusion. Substituting R into (3),

$$R \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}} = \llbracket B \rrbracket_{G_{\mathcal{D}}},$$

which is the correctness condition of Definition 4. $\square$

Theorem 1 gives us a sufficient condition for delegating conjunctive sub-queries in the case where all sources $\mathcal{D}$ are known. However, in discovery-based querying the sources are discovered at runtime, governed by some discovery process which returns during the query execution. As such, determining if we can correctly delegate depends on the *state* of the discovery process, which is defined in Definition 5.

Definition 5 (Discovery state). The state of a discovery setting $W = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc} \rangle$ at time T is the tuple

$$W(T) = \langle \mathbb{D}, \text{data}, \mathcal{D}_0, \text{disc}, \mathcal{D}_d(T) \rangle,$$

the setting W augmented with $\mathcal{D}_d(T)$, the set of sources that data has been called on by time T , subject to

$$\mathcal{D}_0 \subseteq \mathcal{D}_d(T) \subseteq \mathcal{D}_d(T') \subseteq \text{reach}_B(W) \quad \text{for all } T \leq T',$$

i.e. the seeds are discovered from the start, discovery only ever adds sources, and it never adds one outside what disc can reach. For any given $W(T)$ the discovery *frontier* is

$$F(T) = \left(\bigcup_{d_f \in \mathcal{D}_d(T)} \text{disc}(d_f, \text{data}(d_f)) \right) \setminus \mathcal{D}_d(T),$$

the sources that are discovered but do not yet have data called on them. Discovery is complete when $F(T) = \emptyset$.

{TODO: Do we mention Δ still? Stranded paragraph possibly} At any given moment in time T , where discovery has not yet terminated ($\mathcal{D}'(T) \subset \mathcal{D}$), where $\mathcal{D}'(T)$ the set of discovered sources at time t , exclusivity of data sources for set of triple patterns $t_1 \dots t_n$ cannot be determined when $\Delta(t_n, d)$ only exists as a relevance discription for *single* sources.

Definition 6 (Observationally equivalent states). A discovery state $W(T)$ records the whole world, but a query engine at time T has access only to the *observation*

$$O(T) = \langle \mathcal{D}_d(T), \text{data}|_{\mathcal{D}_d(T)} \rangle,$$

the sources it has retrieved together with their contents. Two states $W_1(T)$ and $W_2(T)$ are *observationally equivalent*, written $W_1(T) \sim$

$W_2(T)$, when $O_1(T) = O_2(T)$. Note that $F(T)$ is computable from $O(T)$ and disc: the extent of the frontier is observable, its contents are not.

Theorem 2 (Correct delegation is undetermined under incomplete discovery). Let $W(T)$ be a discovery state for which $F(T) \neq \emptyset$. Then there exist a BGP B , a sub-BGP $B' \subseteq B$, a set of sources $S \subseteq \mathcal{D}_d(T)$, and two states $W_1(T)$ and $W_2(T)$, both observationally equivalent to $W(T)$, such that B' is correctly delegated to S in $W_1(T)$ but not in $W_2(T)$. Since the engine's input is $O(T)$ and all three states share it, no procedure can decide whether the delegation is correct.

Corollary 1 (Delegation requires complete discovery). Correct delegation of B' to S is decidable at time T exactly when $F(T) = \emptyset$, that is, when every source of $\text{reach}_B(W)$ has been retrieved.

PROOF. This requires showing for a g $\square$

From Corollary 1, we find that the problem of runtime federation in discovery-based querying is impossible until all sources are discovered. Depending on the number of sources, which is potentially thousands **{TODO: cite here}**, this is infeasible.

To solve this, we introduce a semantic assumption on what data is considered relevant to our query execution. We call this the *authoritativeness declaration assumption*. In this assumption, we **{TODO: Explain}**. This combines both *authority* who may make statements about a term and *completeness* **{TODO: Explain}**

This assumption is intuitive because **{TODO: Explain intuition with example}**

Formally, we get:

3.3 Delegating Conjunctive Queries

Under this query execution model, we can now investigate delegating queries to composite sources when our model allows determining *correct* delegation under *incomplete* discovery.

3.4 Adaptive Query Planning with Composite Sources

Given the capability of delegating these sub queries, we must now move to the system-side of our problem. Specifically, due to the dynamic discovery of composite sources and the uncertainty of capabilities of these sources, using a static query plan is difficult.

{TODO: Example showing how we can't integrate a sub-query if we have a static plan}

{TODO: Show that even if we use many alternative plans we can still run into issues with overlapping composite sources}

{TODO: Stems!}

3.5 Correctness

3.5.1 Overlapping Sources. Correctness when doing overlapping sources is difficult blabla, cases: Overlapping domain, overlapping triple patterns, both, neither. Given the time dimension, how do we decide what is possible

3.6 Discovery

4 System Application

To evaluate the performance impact of delegating sub-queries we implement hybrid federated link traversal-based query processing. Specifically, within solid blabla

Note this is a concrete application of the stuff introduced here, which is not limited to Solid and link traversal.

5 Evaluation

6 Conclusion

Summarize results and point to future work.

References

- [1] Sijin Cheng and Olaf Hartig. 2020. FedQPL: a language for logical query plans over heterogeneous federations of RDF data sources. In *Proceedings of the 22nd International Conference on Information Integration and Web-based Applications & Services*. 436–445.
- [2] Andreas Schwarte, Peter Haase, Katja Hose, Ralf Schenkel, and Michael Schmidt. 2011. FedX: Optimization Techniques for Federated Query Processing on Linked Data. In *The Semantic Web – ISWC 2011 (LNCS, Vol. 7031)*. Springer, 601–616. doi:10.1007/978-3-642-25073-6_38